UNIFORM DEGREE BOUNDS EXCLUDING BOUNDED OFFSET-SIGNATURE COLORINGS FOR ERDŐS PROBLEM #87

ARGUS OPEN RESEARCH PROJECT

Abstract. Erdős Problem #87 asks whether every graph G with chromatic number $\chi(G) = k$ has Ramsey number $R(G) > c^k R(K_k)$ or satisfies $R(G) \geq R(K_k)$ for sufficiently large k. We prove a uniform maximum-degree bound that excludes the bounded periodic offset-signature construction grammar from producing a small-$R(G)$ k-chromatic witness.

1. Main Bound

Proposition 1.1. *Every graph generated by the bounded periodic offset-signature grammar has maximum degree bounded by a linear function of its signature window width, preventing chromatic number $k \geq 19$ unless the window width scales with k.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1921 in the domain of Ramsey theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped grammar exclusion* deciding the scope: uniform linear maximum-degree bound excluding the bounded periodic offset-signature grammar.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- Erdos Problem #87.

References

[1] Primary problem list and source literature for Result 1921 (Ramsey theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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